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BEFORE YOU BEGIN THE SOFTWARE LOAD - CAUTIONS AND WARNINGS

WARNING

BEFORE STARTING THIS INSTALL THE SYSTEM MUST ALREADY BE AT A SIGNA 8.2.5 SOFTWARE LEVEL OR GREATER IN ORDER TO USE RESTORE INFO INFORMATION. IF YOU ATTEMPT TO RESTORE 8.2 OR EARLIER INFO FILES, THE SYSTEM WILL NOT SCAN DUE TO CALIBRATION AND CONFIGURATION FILE CHANGES.

Important: If your system is connected to a Hospital Network, disconnect it before proceeding. Failure to disconnect the SGI computer can corrupt the software load process.

Important: Disconnect the DASM from the host or remove power from the DASM before beginning the load and remember to reconnect it after the load has finished. The OnLine Center Support has reported intermittent system lockups on Indigo computers during software loads with the DASM connected and from time to time the Octane computer will not load with the DASM running.

Important: During the Install GUI phase of the software load, be sure the **<Num Lock>** key on the key board is turned off. If the **<Num Lock>** is on, the symptom seen is that nothing will happen when trying to type in any of the edit fields in the Install GUI.

Important: When reconfiguring the system using the Install/Configuration GUI, remember to push the **<Configure>** button on the verify tab before going to the Save info tab and doing a SaveInfo. If you have not done the final configuration by pushing the **<Configure>** button in the Verification tab, the information saved during Save Info will not reflect the latest changes made in the Install GUI.

Important: If at any time during the OS load a problem occurs that causes you to stop the install, correct the problem then reset the computer or power cycle the computer and begin the OS install again from the start. This is necessary to insure a complete load. If this does not fix your problem it may be necessary to do a low level format on the system drives. Refer to Appendix C for this process.

Important: If the site will be using the "YP" networking option, do not turn YP on until after the InSite IIP option has been configured and checked out. If YP is turned on first, it will cause the InSite checkout to fail.

Important: If at the end of OS load the system will not boot and stops with the error, "/var/root partition corruption". Do the following:

1. Open a shell window where by typing **sh <Enter>**.
2. Type at the prompt: **dd if=/dev/zero of=/dev/rdisk/dks0d1s0 count=1000 <Enter>**.
This writes zeros across the disk and will run for less than one minute.
3. Start the OS load from the beginning.

This error is a very rare event.

1- LOADING HOST COMPUTER - OVERVIEW & PREPARATION

Note

If this is a new install or system hardware upgrade of a Signa system, the system software has been pre-loaded and will not need to be reloaded. Skip to section 4 for information on how to reconfigure the system for new installs or upgrades.

1-1 Required Media

- IRIX 6.5 Operating System CDROM Set, Disk 1 of 2 and Disk 2 of 2
- IRIX 6.5.4 Boot/Update CDROM
- MrApps - MR Applications CDROM
- MrpResSrv - MR Restricted Service CDROM (*GE Service use only*)
or MrpAdvSrv - MR Advanced Service CDROM (*Licensed In-House Customers*)
- saveINFO - MaxOptix MOD Media (Either blank or previously used)

DO NOT PROCEED UNLESS YOU HAVE ALL THIS MEDIA AVAILABLE

1-2 Archive Images

This installation instruction will erase any current images left in the database. **Archive all images BEFORE continuing with this procedure.**

1-3 Required Information Prior To Loading Software:

First collect vital site information listed below to insure the install will not be stopped for lack of site specific information. To collect this info:

1. On the Service Desktop, select **[Install]**; type **operator <Enter>** for the password.
2. Look in **Hardware Configure**, **Network info**, and **System Configure** buttons for current site information. Refer to Appendix A for instructions on how to locate the site information. Record your Site configuration information in Table 1-1.
 - Unique Internet IP Address for Host (Networked systems only).
 - Unique Internet IP address for PC Computer.
 - Unique Internet IP Address for Laptop port.
 - Netmask
 - TPS Subnet ID
 - Domain Name
 - Hospital Name
 - Hostname (Unique name for this system. Name must begin with Alpha character. Beginning the name with a number will cause the system to not come up.).
 - Unique ID (from GECARES specific to site)
 - Service ID (from GECARES, assigned when first dispatch is opened on system)
 - Suite Name: This name should be at least, but no more than 4 characters.
 - Network Speed
 - Weight Unit

- Field Strength
- Gradient Type
- RF Amp Type
- HIS/RIS Data if option is loaded.
- Coil Type
- Receiver Bandwidth
- System Type (Echospeed, Highspeed, Horizon) *
- Modem Type *
- Camera name *
- DASM type *

* These entries are not found in the Install GUI. They are acquired by physically checking the system.

TABLE 1-1
VITAL SITE INFORMATION

ENTRY	VALUE	ENTRY	VALUE
Host IP:		Host ID:	
Netmask:	(Default is 255.255.252.0)	Unique ID:	
Domain Name:		Coil Type:	__ BRM __ CRM
IWS PC IP:		RF Type:	__ Tube Type RF PEN II __ 1.0T Teli Solid State RF __ 1.5T Analogic Solid State RF
FE Laptop IP:		Service ID:	
TPS Subnet IP:		Weight Unit:	
Network Speed:		Field Strength:	__ 1.0T __ 1.5T
Host Name:	(Name can't begin with a number.)	Gradient Type:	__ 8645 Single or Double Bay __ 8280 SGD Base System __ 8651 SGD HighSlew
Hospital Name:		Camera Type:	
Suite ID:		DASM Type:	__ Digital (LCAM) __ Video (VDB)
Package Name:	__ EchoSpeed SR150 __ EchoSpeed SR120 __ HiSpeed SR77 __ Base (Horizon) SR20	Receiver BW:	__ 62kHz __ 125kHz
HIS/RIS Server IP Address If Option Is Installed			
HIS/RIS Server AE Title			
HIS/RIS Server AE Port #			
MR Server AE Title			
HIS/RIS Patient Name Delimited	[] Yes [] No		

Check (✓) the Options installed at this Site on the System Options Table 1-2 below. After loading software and doing Restore Info, verify that all the options are still available to the customers. If any keys are missing, reactivate them using the option key floppy, or MOD.

TABLE 1-2
SYSTEM OPTIONS DATA TABLE

SITE OPTIONS	SITE OPTIONS
Echo Planar Imaging	Tagging
Fast Gradient Echo	Sgdperf
Cine	DW-EPIEZ
Fast Spin Echo & FLAIR	iDrive Pro
Time of Flight	iDrive
Phase Contrast Vascular Imaging	SmartPrep99 Upgrade
Research	Probe99 Upgrade
Proactive Service	Multi-Nuclear Spectroscopy
Research PSD	Functool
Spectroscopy/PROBE	Voxtool
Videoconferencing	Interavascular Imaging
SGD_EchoSpeed	Clariview
DW EPI	Multislice Multiangle
FLAIR EPI	Spectro Analysis GE
SPECIAL	
Smart Prep	
SSFSE	
Three Plane Localizer	
HIS RIS	
E3D TOF	
FSE_XL	
Bloodsupp	
Fastcine	

1-4 SaveInfo and Saving System Information Entered In The Install GUI

1-4-1 SaveInfo

A SaveInfo should be done before starting the install to save the latest system configuration files, calibration files, and protocols.

1. Insert the your SaveInfo MOD or a new MOD into the drive.
2. From the Service Desktop select **[Install]**. In the Install Tool window, type **operator** **<Enter>** at the password prompt.
3. If existing software is release 8.2.5
 - a. Click on **[Save Info]** from the Install menu. The system opens the window called Xexec. (If the MOD has not been used, you will see a message regarding the use of the MOD for "Raw Data Storage"; chose **[Overwrite]**).
 - b. When asked: Do you really want to save configuration files to the Info MOD? [y|n], type: **y <Enter>**.
 - c. When SaveInfo is complete, remove the MOD media and write the date on it.

If existing software is Release 8.3 or 8.4

- a. In the Install GUI, select the Save/Restore tab.
- b. There are four options: **[Save Information]**, **[Save Protocols]**, **[Restore Info]** and **[Restore prot]**. To save all information, select **[Save Information]**; this saves all system calibration and configuration files, along with customer protocols. **The [Save/Restore Protocol]** option is intended as a means to quickly save just customer protocols on the same or different MOD.
- c. When SaveInfo is complete, remove the MOD media and write the date on it.

1-4-2 Save GI Configurations (Release 8.3 M3 and Later)

If you are loading software upgrade on a system already at 8.3_M3 or later software, create a GI (Guided Install) configure MOD from the Guided Install File menu.

1. Insert the your normal SaveInfo MOD or a new MOD into the drive.
2. At top left corner of the Install GUI, select **<File>** then **<Save GI Configuration to MOD>**.

Note

The "save to MOD" process does not create a complete SaveInfo disk. It only creates a copy of the information already entered in the GUI tabs for later use to update the GUI tabs during a new install. (*SaveInfo uses the Save/Restore tab.*) The **<Save GI Configuration to MOD>** and SaveInfo can be saved on the same MOD. A separate partition is built for each.

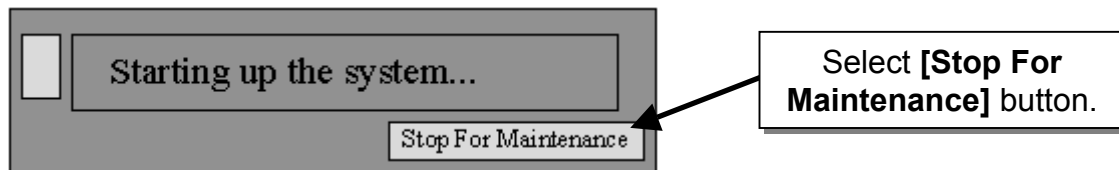
2- HOST OPERATING SYSTEM SOFTWARE LOAD

Initial Conditions:

- All images archived (Section 1-2).
- System information recorded (Section 1-3); SaveInfo performed (Section 1-4).
- System disconnected from Hospital Network.
- Indigo Host Computer: DASM disconnected from Host.

Procedure:

1. If the system is not up and running, switch on power at this time and continue with step 2. If the system is already running, go to the Service Desktop and select **[Shutdown]**. When prompted, click on the **[Restart Button]**.
2. Stop the reboot process; click on the **[Stop For Maintenance]** button when it appears.



3. Click on the **[Enter Command Monitor]** button when it becomes available.
4. Insert the **IRIX 6.5** CD labeled "**Boot/Update**" into the CDROM Drive on the workstation (right side). On the command line type the following (*user responses are bolded text*):

Depending on the host computer type your system is using, boot the CD by typing one of the following:

For SGI Octane computer:

```
>> boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 --x <Enter>
```

For SGI Indigo computer:

```
>> boot -f dksc(0,3,8)sashARCS dksc(0,3,7)stand/fx.ARCS --x <Enter>
```

Note

The rest of the install is for all SGI computer types.

```
fx: "device-name" = (dksc) <Enter>
```

```
.
```

```
fx: ctrl# = (0) <Enter>
```

```
.
```

```
fx: drive# = (1) <Enter>
```

```
.
```

```
fx> r/ro <Enter>
```

```
.
```

```
fx/Repartition/RootDrive: Type of Data Partition = (xfs) <Enter>
```

```
.
```

```
fx> continue? yes <Enter>
```

.
fx> **l/sy** <Enter> (This is a lower case l , /, lower case s, and lower case y.)

.
fx> **exit** <Enter>

5. At this point the system will reboot to the PROM startup screen. Select **[Install System Software]** button. A pop up window opens asking where to install from. Choose, **[Local CD-ROM]**, select **[Install]**. Then select the **[Continue]** button.

If you are working with newly formatted disk drives or new, never used disk drives, you will be told by the system during this boot that the system was unable to mount the "/root" file system and the system will stop with the statement "press Enter to invoke C shell csh:". If you see this message, type the following:

```
...press Enter to invoke C shell csh: <Enter>
# mkfs /dev/dsk/dks0d1s0 (This is /dev/dsk/dks zero d one s zero.)
# exit <Enter>
```

6. The next system message should be "*Trying again to mount /dev/dsk/dks0d1s0 on /root.*" If you get this message, continue with step 7. Otherwise...

If you are working with a newly formatted drive you may get a message similar to:
/dev/dsk/dks0d1s0: Invalid argument No valid file system
found on: /dev/dsk/dks0d1s0 This is your system disk:
without it we have nothing on which to install software.

Do the following:

```
Make new file system on /dev/dsk/dks0d1s0 [yes/no/sh/help]: yes<Enter>
About to remake (mkfs) file system on: /dev/dsk/dks0d1s0.
This will destroy all data on disk partition: /dev/dsk/dks0d1s0.
Are you sure? [y/n] (n): y<Enter>
```

```
Block size of filesystem 512 or 4096 bytes? 4096<Enter>
Doing: mkfs -b size=4096 /dev/dsk/dks0d1s0
meta-data=/dev/dsk/dks0d1s0 isize=256 agcount=8, agsize=128612 blks
data = bsize=4096 blocks=1028895, imaxpct=25
log =internal log bsize=4096 blocks=1000
realtime =none bsize=65536 blocks=0, rtextents=0
Trying again to mount /dev/dsk/dks0d1s0 on /root.
```

7. Continue as follows:

```
inst> sh <Enter>
```

```
.
# mount /CDROM <Enter>
```


/CDROM/mr_prep <Enter>

.

8. The system will run for about 6 minutes and then will prompt with:

"Please insert the First OS Disk (Pa) and then press <enter>".

Put the OS CDROM Disk 1 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

9. After an additional 3-6 minutes, the system will again prompt with:

"Please Insert the Second OS Disk (Pb), then press <enter>"

Put the OS CDROM Disk 2 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

Loading the rest of the Operating System off the second disk will take about 30 minutes and will not require any user intervention.

10. When finished, the system will stop at the "#" prompt. Type the following to continue.

.

exit <Enter> (to exit back to the original inst> prompt.)

.

.

Inst> **exit** <Enter>

.

Automatically reconfiguring the operating system..

.

.

.....Ready to restart... **y** <Enter> (Restarts the system)

.

.

The system will finish some processing and reboot. The OS load portion of the Software Install is complete.

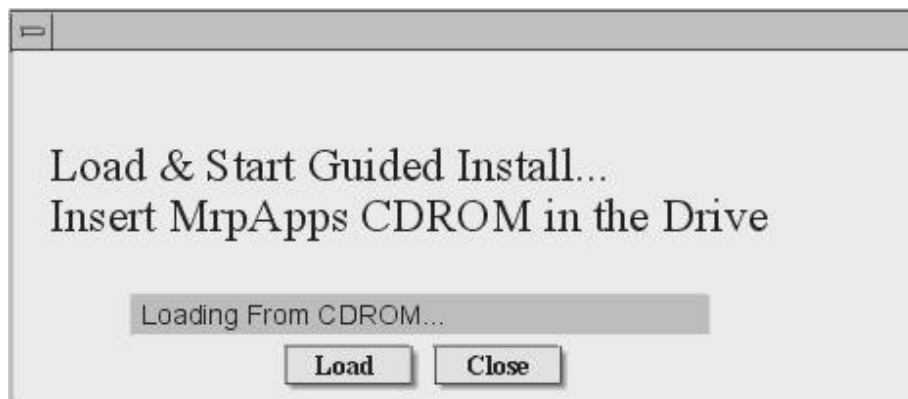
Important!

Do not log into EZ Setup. It is no longer used as part of Install.

11. Continue to Section 3 of this document for the MR_Apps software load.

3- MR_APPS SOFTWARE LOAD

1. When the system is finished rebooting, log in as **root**. There is no password at this time. You will first see a startup box that will ask for the MrpApps CDROM. See Illustration 3-1.



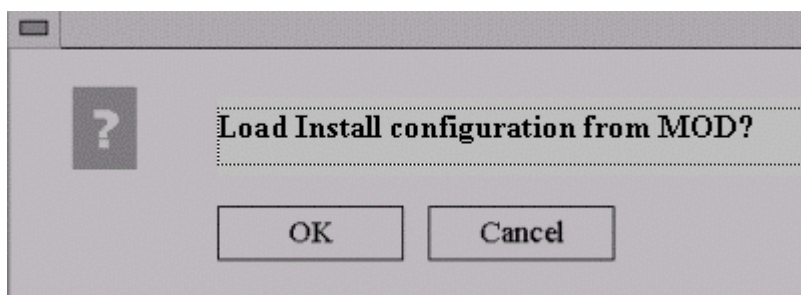
GUIDED INSTALL LOAD SCREEN
ILLUSTRATION 3-1

2. Remove any CD from the CDROM drive and insert the MrpApps CDROM into the CDROM drive. Wait 20 seconds or until the amber light on the CDROM drive stops blinking and push the **<Load>** button. The Guided Install will load and start.

Note

Help documentation is built into the Guided Install GUI. Push the **<Help>** button on the tab you need information for in the GUI. When the GUI first starts, press the **<What's New>** button for information on using the GUI.

3. On initial startup, a popup box will appear over the Guided Install interface describing the tool and its use. After reading the information push the **<OK>** button to continue.
4. Next, a popup box (see Illustration 3-2) will appear over the Guided Install interface.



INSTALL CONFIGURATION POPUP SCREEN
ILLUSTRATION 3-2

If you are reloading 8.3 or later software, insert the SaveInfo MOD (created in Section 1-4-2, Save GI Configurations) then push the **<OK>** button to reinstall your last GUI install configuration values. *(The normal SaveInfo files are not restored at this point in the software load. They will be restored in step 9.)* You will verify the correct values were restored by checking each tab in the next step.

Important: The last Install GUI configuration can only be installed if it was previously saved using the **<Save GI Configuration to MOD>** option per Section 1-4-2, Save GI configurations.

If you did not perform Section 1-4-2, Save GI Configurations (for example, sites upgrading from 8.2.5 software), push the **<Cancel>** button to continue manually with the install.

5. The Guided Install home screen is displayed. See Illustration 3-3.

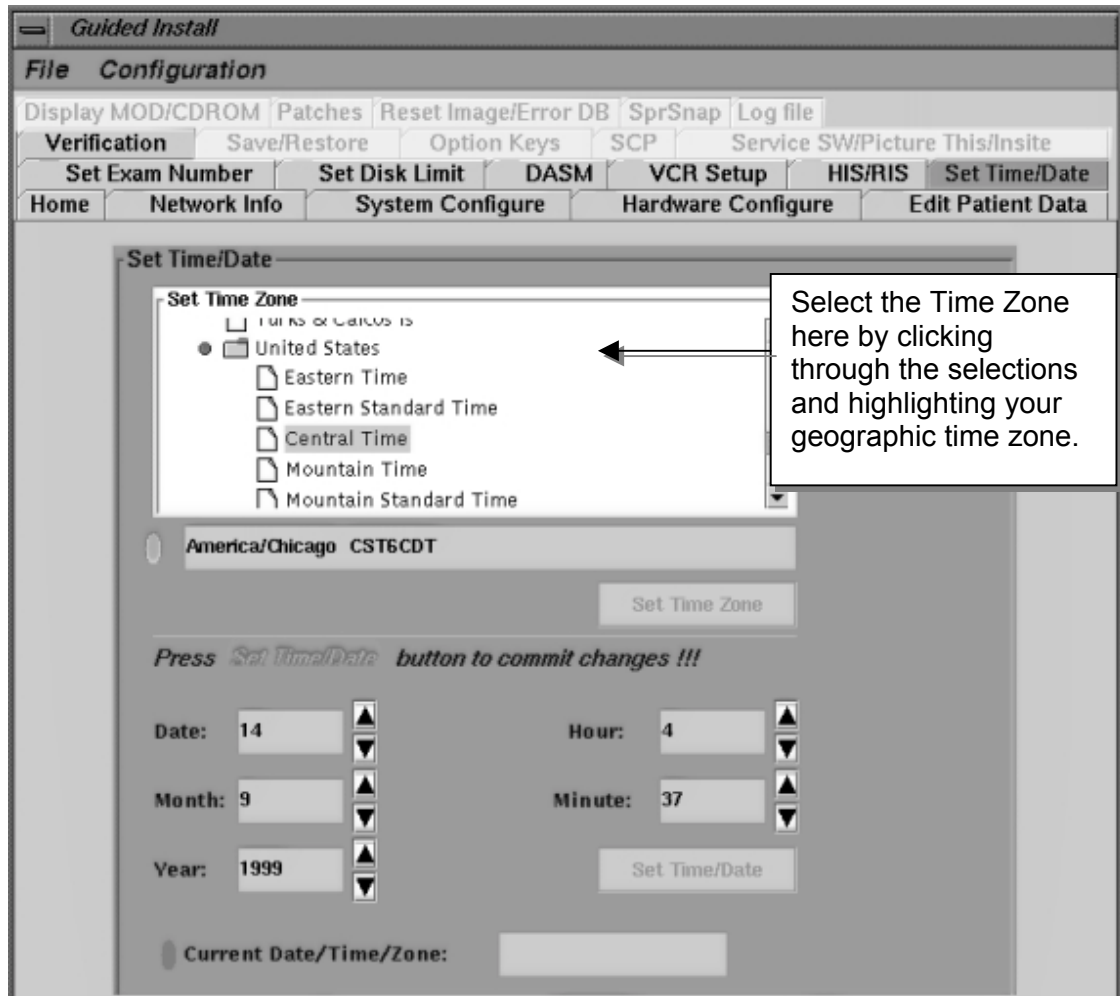


GUIDED INSTALL - HOME SCREEN
ILLUSTRATION 3-3

6. Press the **<Next Tab>** button to move through active tabs from **“Network Info”** to **“Set Time/Data”**. Fill out all empty entries or make any updates as appropriate. This information should have been saved in Table 1-1 of this document. When you reach the “Set Time/Date” tab, go to the “Set Time Zone” box and use the selections to set the time zone your system is in. See Illustration 3-4.

Note

If YP is turned on, the system will take longer to boot (2 to 4 minutes). Turn YP on only if the site's external network requires it. Be aware there is a small delay that may not be seen on other sites.

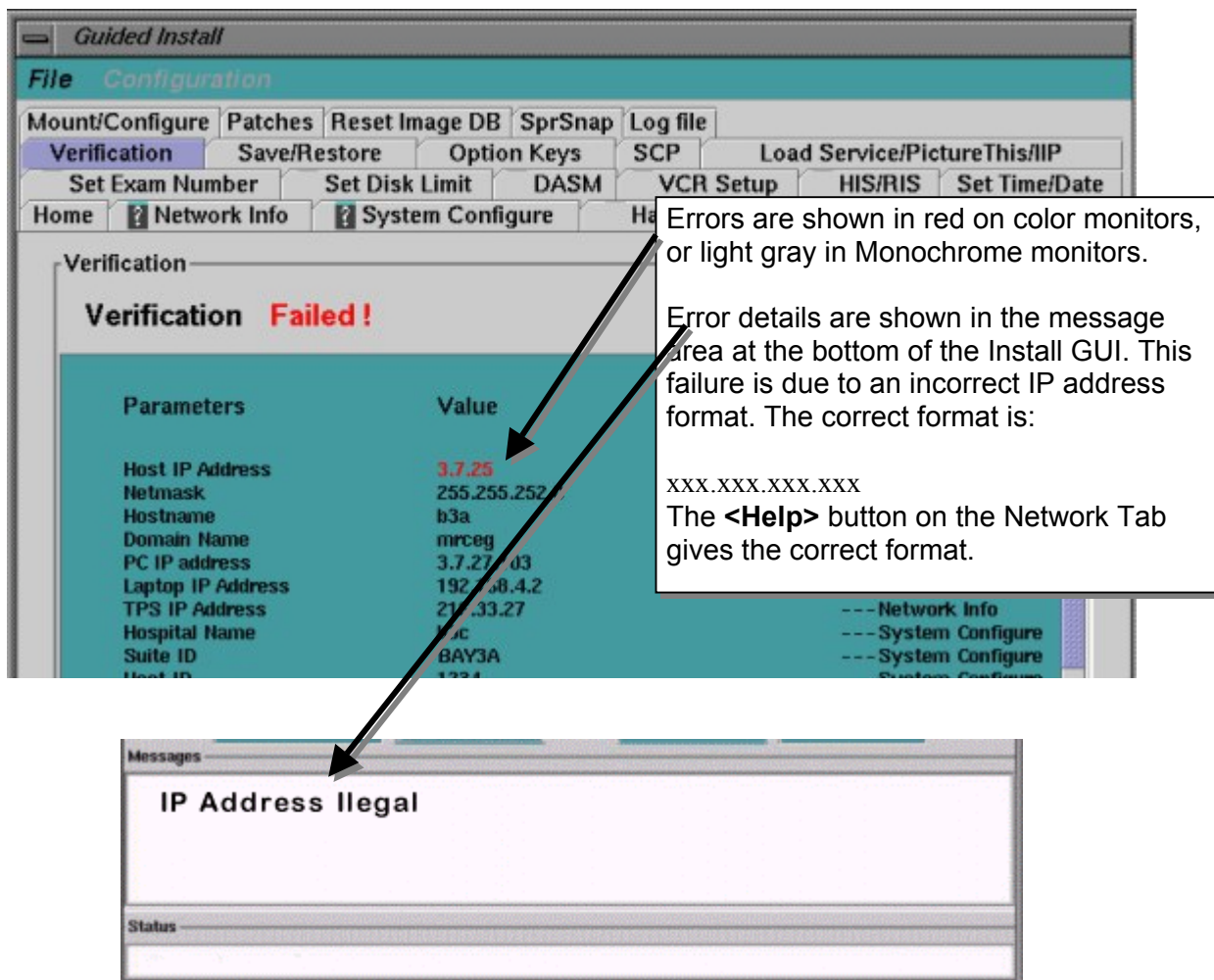


GUIDED INSTALL - SET TIME/DATE SCREEN
ILLUSTRATION 3-4

7. Proceed to the **Verification** tab. Once in the Verification tab, look at the pass or fail status of the entries. If any entry has a failed status, the **<Install Software>** button will not be active. Failed entries will be due to missing entries, or entries not in the correct format. Go to the tab with the error and correct it, then push the **<Install>** button to open the Verification tab again. If all entries pass verification at this point, push the **<Install Software>** button again to begin the MR Apps load. The MR Apps load will take 20-30 minutes, depending on the machine being loaded.

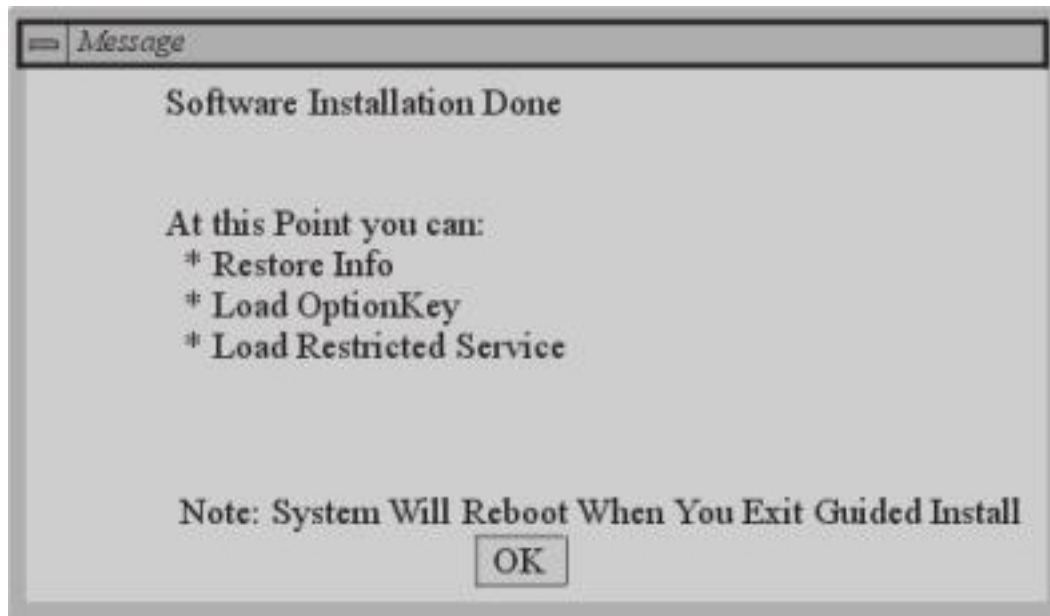
Note

It is not necessary that any one tab be filled out in any particular order. Just all required configuration tabs must be filled in or the install will not continue. The Verification Tab will list any formatting errors as well as current settings. See Illustration 3-5 for an example of a failure in the Verification Tab in networking (in the example, the Host IP Address is an incorrect format).



GUIDED INSTALL - VERIFICATION SCREEN
ILLUSTRATION 3-5

8. Once the MR Apps load is done the popup in Illustration 3-6 is displayed. Push the **<OK>** button to close the popup.



GUIDED INSTALL – SW DONE SCREEN

ILLUSTRATION 3-6

9. At this point the GUI will change from the Install phase to the Configure phase. If a SaveInfo info MOD is available for this site, proceed to the **Save/Restore** tab and Restore the system info. If no SaveInfo MOD is available, go to each tab in the Install GUI and fill in or update all system information manually. A help button is available on each tab, which will open a popup box with information on the required data and its format for that page.
10. When the RestoreInfo is finished a popup will open telling the installer that upgrade scripts must be run. See Illustration 3-7.



GUIDED INSTALL – UPGRADE SCRIPTS MESSAGE

ILLUSTRATION 3-7

11. Push the **<OK>** to continue. The upgrade script menu will open (see Illustration 3-8).

This script upgrades the system config files For LFC software. After Restore Info is completed, the following UP scripts must be run before scanning.	
1. ConfigUP	Update MRconfig.cfg/GradientConfig.cfg files
2. CoilUp	Update CoilConfig.cfg file
3. Protocol1Up	Upgrades 8.2.5/83 Protocols
4. ProtFormat	Reformats site protocols names
5. Restore_License	Must be run individually
6. Run all scripts	Except Restore_License
7. Exit	Return to Guided Install.

GUIDED INSTALL – UPGRADE SCRIPTS MENU

ILLUSTRATION 3-8

You must always run these updates in the following order:

- a. Type: **6 <Enter>** *This will take about 1 minute to run.*
- b. Type: **5 <Enter>** *This will take < 1 minute to run.*
- c. Type: **7 <Enter>** *This will close the menu and return to the GUI.*

Note

Refer to Appendix A for parameters updated by the Configup scripts

12. Once back on the Guided Install screen, use the cursor to select **<File>** and then **<Quit>** at the top left of the GUI. When the GUI shuts down, the system will reboot and all changes restored or made will be set.
13. After the system has finished rebooting, once again log in as **root**. Password is **operator**.
14. You will be asked, once entering the root desktop, if you want to start the Install GUI. Push the **<Yes>** button.
15. When the install GUI opens, go to the **Load ServiceSW/Picture this/InSite** tab and install service software. *(Restricted Service Software is only available for GE use; Advanced Service Software is only available to sites with a valid Advanced Service Package Limited License.)* GEMS Field Engineers loading GEMS Restricted Service Software should also configure InSite Interactive at this time (see Illustration 3-9). Follow the instructions given in popup windows as presented.

IIPConfig X

InSite Interactive Platform Configuration

Information ProDiags Health Page **Device Connection** InSite Checkout

Connection Device Configuration

Last Update: Fri Aug 13 15:50:42 CDT 1999

Device Connection Type Modem ▼

Select the InSite Connection Type above, or select the Modem settings given below and Press

APPLY – To update the modem and modem configuration files with model and country defaults.

CUSTOM – For other modem settings.

Dial-out Prefix 8, ▼ Dialing mode Tone ▼

Modem Type USRobotics Courier V.34 or V.Everything ▼ Country Default – All Others ▼

CPU Serial Port Name /dev/ttyf1 ▼ CPU Serial Port Speed 115200 ▼

APPLY CUSTOM >>

Status: IIP Configuration Tool Ready for Input... EXIT

CONFIGURE INSITE MENU
ILLUSTRATION 3-9

16. After service software installation, move through and check the accuracy of each tab. Insure all system options for the site are active. If needed, load software option keys at this time. If the site has any of the following option keys, they must always be reloaded, as they are not saved with SaveInfo: *IVI, Voxtool, FuncTool, Clairview*.

Note

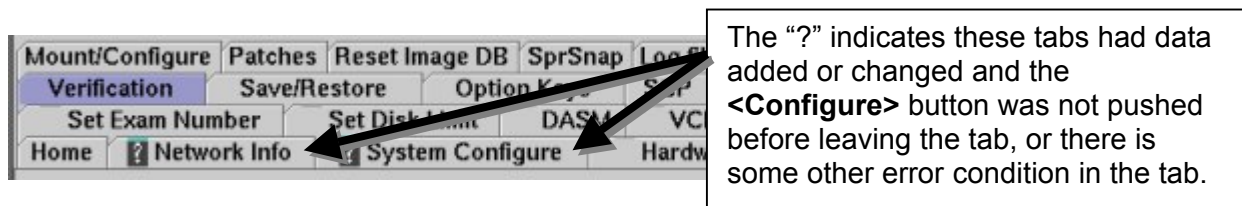
When configuring the System Health page of **Configure InSite Menu**, you are able to enter multiple email address where reports are sent. The instructions on that tab are incomplete. You must push the **<Enter>** key after every email address entered.

Note

Release 8.3 version M4 and later software comes with two software MOD keys that must always be reloaded. These two MOD's (two sides on one MOD and one side on the other MOD to load) contain the LX Tools software that is shipped with every system.

17. Turn YP on only if the site's external network requires it. If YP is turned on, the system will take longer to boot (2 to 4 minutes). Be aware there is a small delay that may not be seen on the sites.
18. After information as been restored from MOD or changed in a tab, and configuration updates have been run, push the **<Configure>** button on the bottom of the tab to apply the configuration.

If the **<Configure>** button is not pushed to commit changes to a tab, or an error is present in a tab, a question mark will be seen on the tab, as shown in Illustration 3-10.



CONFIGURATION PROBLEM EXAMPLE

ILLUSTRATION 3-10

18. Perform a SaveInfo to save the latest changes by inserting the site SaveInfo or blank MOD into the drive then going to the **Save/Restore** Tab and pressing the **[SaveInfo]** button.
19. Insert the your normal SaveInfo MOD or a new MOD into the drive if not already in the drive.
20. At top left corner of the Install GUI, select **<File>** then **<Save GI Configuration to MOD>**. This will save information that has been entered in all the edit fields in the GUI.

Note

The “save to MOD” process does not create a complete SaveInfo disk. It only creates a copy of the information already entered in the GUI tabs for later use to update the GUI tabs during a new install. (*SaveInfo uses the Save/Restore tab.*) The **<Save GI Configuration to MOD>** and SaveInfo can be saved on the same MOD. A separate partition is built for each.

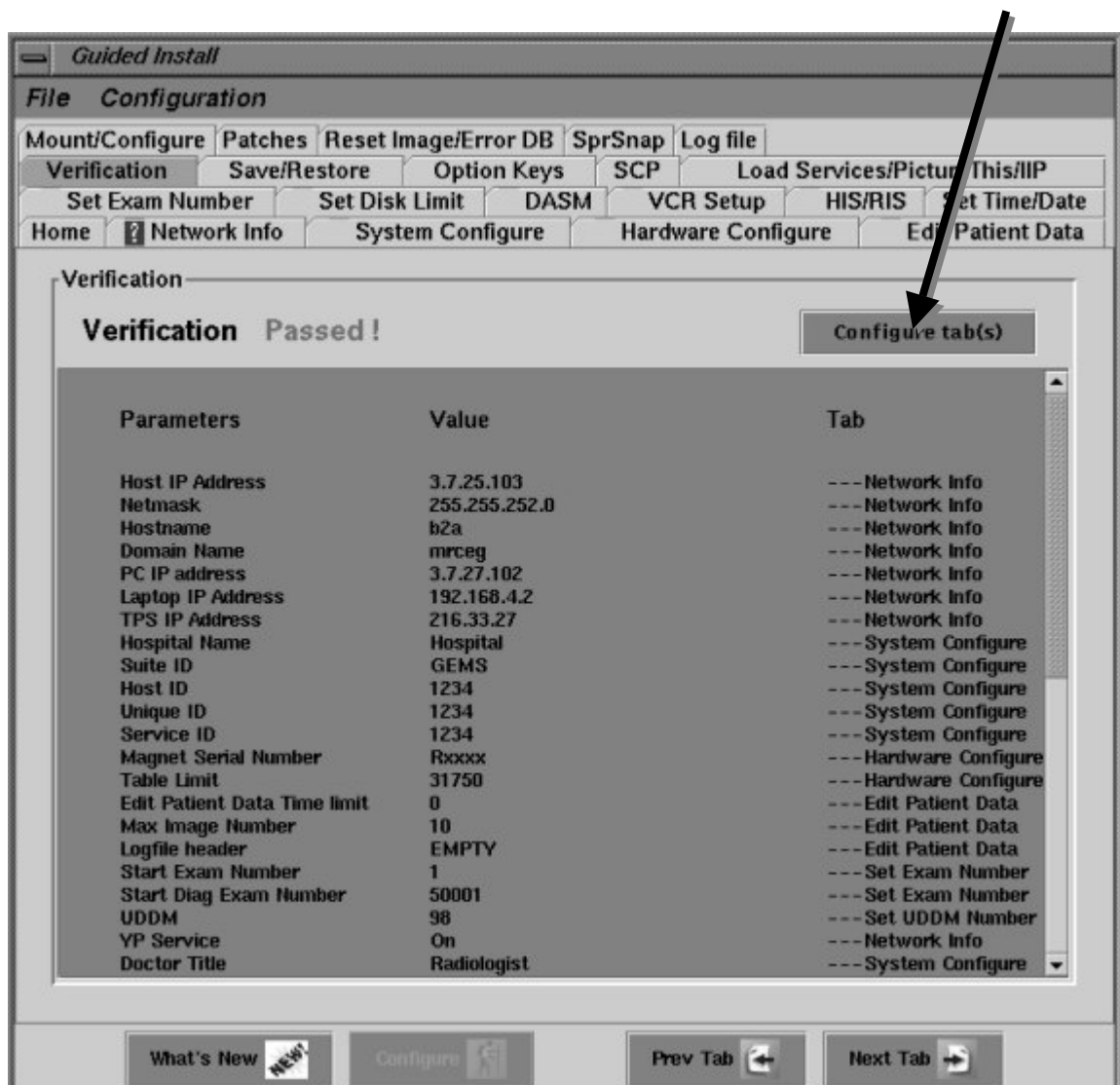
21. Exit the install GUI by selecting **<File>** then **<Quit>** from the top of the interface. If any error conditions still exist, you will be warned that no changes will be made. If no error conditions are found, the GUI will save user parameters.
22. Reconnect the following hardware at this time if it was disconnected at the beginning of the install:
 - Network Interface
 - DASM
23. Log out of root and reboot the system. Log into Signa.

24. This concludes the software load. Continue with Section 5 to update the Monitor gamma settings.

4- RECONFIGURING THE SYSTEM USING THE INSTALL GUI

If this is a new/upgraded Signa installation or if the system just needs to be re-configured between software loads, do the following to start the Configuration GUI:

1. With Signa up and running, go to the service desktop and push the **<Install>** button on the toolbar. A C-Shell will open and ask for root password ("**operator**" **<Enter>** is the default). The Install GUI will load and start.
2. Help documentation is built into the GUI. Push the **<Help>** button on the tab you need information for in the GUI. Push the **<Next>** button to move through tabs in order. Modify entries to match your system as required. When satisfied, push the **<Configure>** button at the bottom of the tab page or wait until all tabs have been modified and go to the "**Verification**" tab. Review changes made are correct, then push the **<Configure Tab(s)>** button to update the system (see Illustration 4-1).



VERIFICATION MENU – CONFIGURE TABS
ILLUSTRATION 4-1

3. Before exiting the GUI, insert the SaveInfo MOD or new MOD into the drive. Go to the top left corner of the Install GUI and select **<File>** and then **<Save GI Configuration to MOD>**. This process does not create a SaveInfo disk. It just creates a copy of the information already entered in the GUI tabs for use in the next software install.
4. Exit the install GUI by selecting **<File>** then **<Quit>**. If any error conditions still exist, you will be warned that no changes will be made before exiting.

Note

The GUI will not automatically reboot the system in configure mode. It will be up to the user to reboot after making configurations.

5. If changes were made to the system configuration, select **[System Shutdown]** and reboot the systems for changes to take effect.
6. For new system installations or upgrades with preloaded software, continue with Section 5, Monitor Gamma Value Reset.

5- MONITOR GAMMA VALUE RESET

After a complete software load has finished, do the following steps to update the monitor configurations. This insures compatibility between monitor and camera.

1. In the C-Shell, type **su <Enter>** to log in as root; enter the root password (default is **operator**).
2. **B/W Monitor:**
 - a. Type **jot /usr/g/grayPix/grayPix.cfg <Enter>** to edit the grayPix.cfg file.
 - b. Modify the numerical value for "gamma" to **0.9**

Color CRT Monitor:

- a. Type **jot /usr/g/colorPix/colorPix.cfg <Enter>** to edit the colorPix.cfg file.
- b. Modify the numerical value for "gamma" to **1.1**

Color LCD Monitor:

- a. Type **jot /usr/g/colorPix/colorPix.cfg <Enter>** to edit the colorPix.cfg file.
- b. Modify the numerical value for "gamma" to **1.05**

3. In the jot window, select **File -> Save**, then **File -> Exit** to save the changes. This modifies the gamma value so whenever the system reboots, the new value is not lost.
4. In the C-Shell, enter one of the following:

B/W Monitor: **gamma 0.9 <Enter>**

Color CRT Monitor: **gamma 1.1 <Enter>**

Color LCD Monitor: **gamma 1.05 <Enter>**

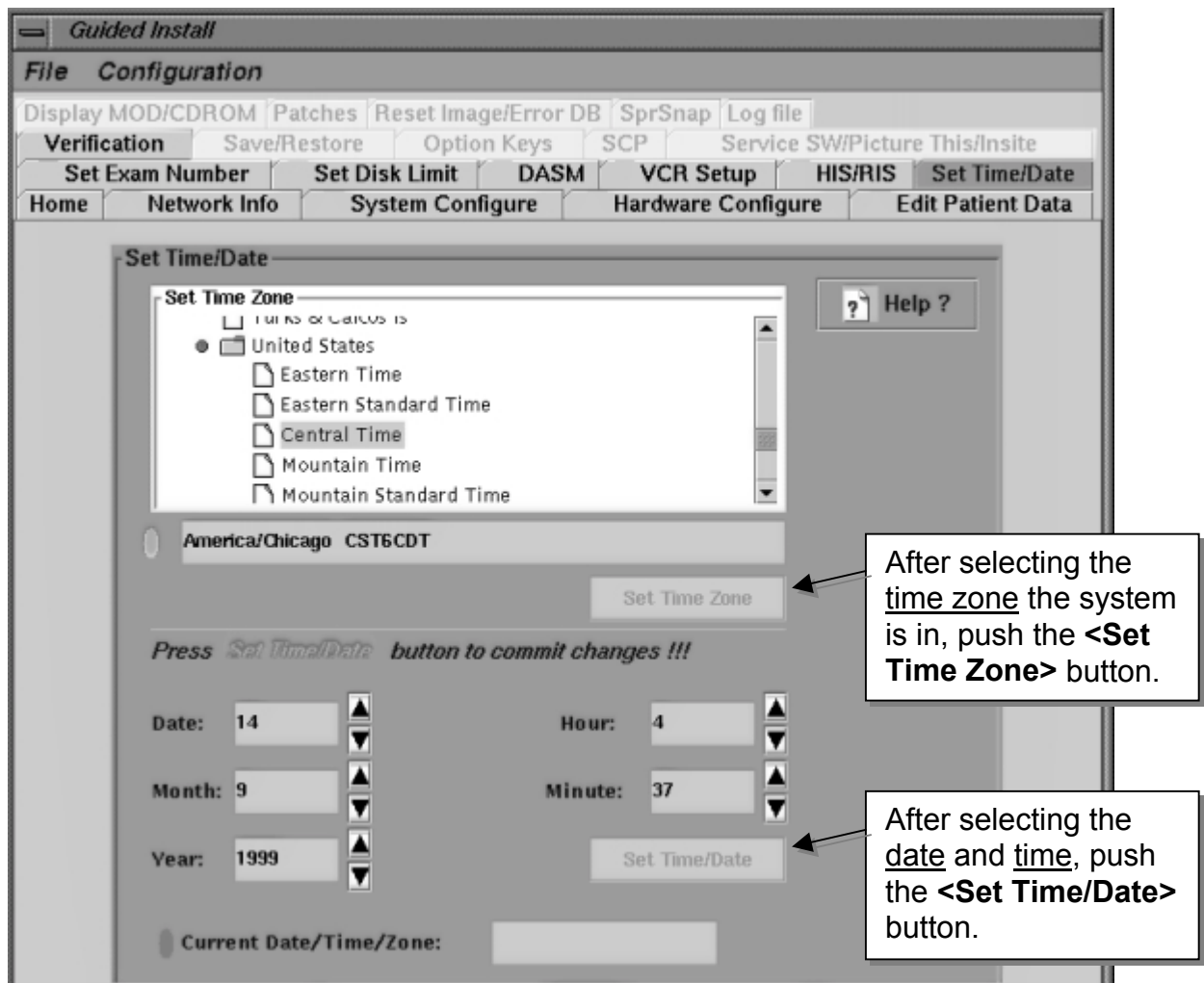
This is in addition to steps 2 and 3. This step immediately changes the gamma value without the need to reboot the system.

5. If you performed a complete software load, you are now done. For new system installations or upgrades with preloaded software, continue with Section 6, Set Time/Date Tab to complete the software reconfigurations.

6- SET TIME/DATE TAB

The **Time/Date** tab is the only tab that the Configure button will never be highlighted to set options. There are separate buttons for configuring this tab (see Illustration 6-1). If you need to re-configure the Host time and/or date at any time, do the following:

1. Select the Time Zone your system is located in, then push the **<Set Time Zone>** button.
2. Select the date and local time, then push the **<Set Time/Date>** button.



SET TIME/DATE MENU
ILLUSTRATION 6-1

APPENDIX A - HOST SOFTWARE INSTALL WORKSHEETS

Host IP Address: The Host IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

IWS PC IP Address: The IWS PC IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

FE Laptop IP Address: The Laptop IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

TPS Subnet Address: The TPS is on a subnet of the Signa LX scanner. A default address is loaded during the load from cold process and it should not be changed.

Hostname: This is the network name given to the scanner host computer.

Hospital Name: The Name of the hospital or other facility where the Horizon LX is installed. Name can include characters and/or integers up to 32 characters.

Doctors Title: The title to be used at the scanning site by the reading physician (Radiologist or Diagnostician).

Suite ID: Name of the Suite (if any) the scanner will be connected to. Leave blank otherwise.

Unique ID: The Unique System ID is a 16 alphanumeric character string assigned at the time the system is first installed. The Unique system ID is assigned by GE CARES when the first dispatch is opened to document the start of installation.

Service ID: The Service ID is a 16 alphanumeric character string assigned at the time the system is first installed. This is the same number that GE CARES uses to track financial and administrative data for the system.

Weight Unit: The weight system to be used by the scanner. Kilo Grams or US pounds.

Monitor type: The type of monitor the scanner will be using.

Windows 95 OEM Number: The Microsoft OEM Number found on the "Proof of Authenticity" for Windows 95. _____

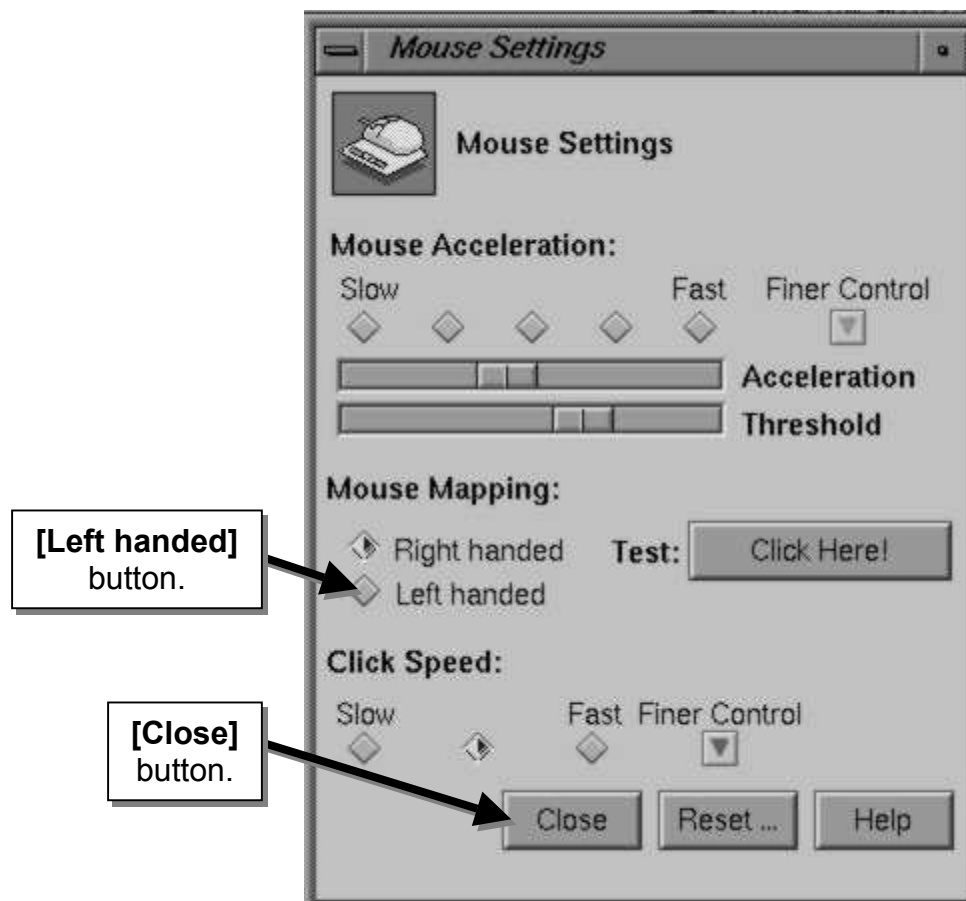
NetMask: For new installs, the Netmask should be obtained from the site network administrator if the scanner will be connected to the site's internal network. If this is a stand alone system, use the default netmask. For Release 8.3 and later, the netmask value is available from the Install GUI. For Release 8.2.5 systems, do the following to obtain the netmask value: **[System Shutdown]**, Log in as Root, type **operator <Enter>**, select **[Reconfig]** then **[Basic Network Setup]** to view the netmask.

APPENDIX B - CONFIGURING MOUSE FOR LEFT HANDED USE

The system Host mouse is default configured for right handed use. It can be configured for permanent or temporary left handed use.

B-1 Configuring Mouse For Permanent Left Handed Use

1. If Signa is up and running, you must first log out and bring the system up to the Login screen. Otherwise, turn the system on and proceed to the Login screen.
2. Log in as **root**. The system default password is “**operator**”. On the root desktop, use the mouse cursor on the ToolChest and click and slide through **[Desk Top]** then **[Customize]** then **[Mouse]**.
3. Push the **[Left Handed]** button under Mouse Mapping, then push the **[Close]** Button to accept the changes. See Illustration B-1.



SET TIME/DATE MENU
ILLUSTRATION B-1

4. Log out of the root desktop and reboot the system. To change the mouse mapping back to Right handed. Repeat this section but push the **[Right Handed]** button, then reboot the system.

B-2 Configuring Mouse For Temporary Left Handed Use

Using this method will change the mouse mapping from right handed to left, but only until Signa is rebooted, when it will revert back to right handed mapping.

1. If not already at the Signa scanning level, go there. Open a command window and on the command line type: **mouse <Enter>**.
2. Push the **[Left Handed]** button under Mouse Mapping, then push the **[Close]** Button to accept the changes. (See Illustration B-1.)
3. To reconfigure the mouse back to right handed operation, repeat this section but push the **[Right Handed]** button, or reboot Signa.

APPENDIX C - DISK DRIVE LOW LEVEL FORMAT

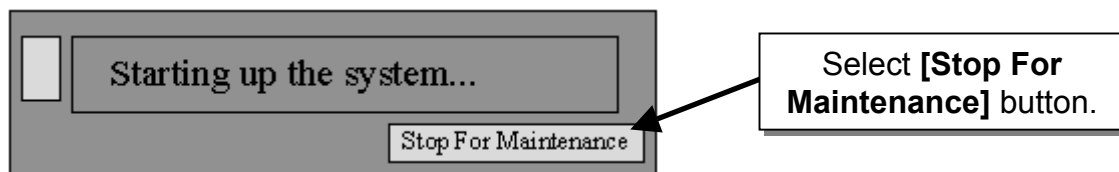
This section describes how to do a low level format of the disk drives. A low level format should only be done if loading a different Operating System (OS) or if suspected bad blocks on a disk drive are causing system problems. Depending on the system and disk drives, this procedure can last up to 1 hour per drive or as little as 20 minutes per drive.

1. Insert the **IRIX 6.5** CD labeled "**Boot/Update**" into the CDROM Drive on the workstation (right side).

Note

User responses are bolded text

2. With the system up and running, go to the service desktop and select **[Shutdown]**. If the system is not up and running, switch on power at this time.
3. When prompted, click on the **[Restart Button]**. Stop the reboot process by clicking on the **[Stop For Maintenance]** button when it appears.



4. Next, click on the **[Enter Command Monitor]** button when it becomes available. Type one of the following depending on the system your working on:

For SGI Octane computer type:

```
>> boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 --x <Enter>
```

For SGI Indigo computer, type:

```
>> boot -f dksc(0,3,8)sashARCS dksc(0,3,7)stand/fx.ARCS --x <Enter>
```

5. The rest of the install does not depend on the type of SGI computer (Indigo/Octane). Continue typing as follows:

```
fx: "device-name" = (dksc) <Enter>
```

```
.
```

```
fx: ctrlr# = (0) <Enter>
```

```
.
```

```
fx: drive# = (1) <Enter> or 2 <Enter> if doing the second drive.
```

```
.
```

```
fx: d/fo <Enter>
```

```
.
```

```
Press <Enter> to accept current parameters.
```

```
.
```

```
Type: Yes <Enter> to start the format.
```

6. The format can take up to 1 hour, depending on the system and disk drives that are being formatted. There will be no prompts to the user while the format is running. Be patient and do not interfere with the system. When the format is finished the system will prompt with:

Format completed successfully

.

.

fx/debug>

7. Reset or power cycle the computer to exit debug and proceed to Section 2 of this document to proceed with the OS (Operating System) load for the system.

APPENDIX D - GENERAL INFORMATION AND NICE TO KNOWS

D-1 Bore Temperature Sensor - Must Be Working!

If the bore temperature sensor is disabled, sites with iDrive software will have problems. Sites running the iDrive software must have a functional bore temperature sense circuit. The symptom for a non-functional Bore sensor will be evident when an iDrive scan is attempted; the prescan will complete OK, but the scan will stop immediately after the first pulse is heard in the body coil. No images will be produced and no errors will appear in the log.

If the system has iDrive software (as listed in *Option Key* tab of Guided Install), to verify functional bore temperature circuit, set up the scan in Table D-1 (or any other Real Time scan). Download the scan and select **[Scan]**. If prescan completes and the scan continues to run, you have a functional bore temperature circuit. If prescan completes OK, but the scan stops immediately after the first pulse, check the bore temperature circuit.

TABLE D-1
IDRIVE SAMPLE PROTOCOL

<u>PATIENT REGISTER</u>	[New Pt]	<u>SCANNING RANGE</u>	
<u>PATIENT INFORMATION</u>		FOV.	[32]
Patient Id	geservice	Slice Thickness.	[7]
Patient Name	test	Spacing	[1.5]
Weight (Lb)	111	Start	0
	[Landmark]	End..	0
Landmark	[Sternal Notch]	# Slices	1 (default)
		L/R Center	0 (default)
<u>PATIENT PROTOCOLS</u>	[Patient Position]	P/A Center	0 (default)
		Table Delta	0.00 (default)
<u>PATIENT POSITION</u>		<u>ACQUISITION TIMING</u>	
Patient Position	[>] [Supine]	Freq	[128]
Patient Entry.	[>] [Head First]	Phase	[128]
Coil	[...] [Body] [Accept]	NEX	[1]
<u>IMAGING PARAMETERS</u>		Phase FOV	1.00 (default)
Plane	[>] [Axial]	Freq Dir	[>] [R/L]
Mode	[>] [2D]	Auto Center Freq	[>] [water]
Pulse Seq	[...][Fast GRE] [Accept]	Autoshim	[on]
Imaging Options	[Real-time]	(lowest window)	[Save Series]
Psd Name	no entry		
Protocol	no entry		
<u>SCAN TIMING</u>			
# of Echoes	1 (default)		
TE	[Min Full]		
TR.	[Minimum]		
Flip Angle	[20]		
Bandwidth	[62.50]		

D-2 Signa Power up Delay

When Signa is first powered on there is a considerable period of time that nothing is seen on the monitor; up to 4 minutes depending on the computer type. The system is checking configurations and starting software processes at this time. If you are concerned that the system is hung, push the <Esc> button on the top left side of the keyboard. The system will then post to the monitor all its boot up activities.

D-3 "Configup" Script Modifications To Files

When RestoreINFO is run after upgrading from Release 8.2.5 software the first time, new version configuration files are overwritten. The "configup" script updates the MR System configuration file with the new changes as needed:

1. The "configup" script updates the MRconfig.cfg file with the new "fastprescan" parameter if it is not present (if "fastprescan" parameter is present, it does not add or change it):

```
# This field is added by shivar@wge for MRIge47785
# A fastprescan field to turn Faster Prescan on & off .
#           type = integer, legal values = 0 (off)
#                                           = 1 (on)
fastprescan = "1"
```

2. The "configup" script updates the MRconfig.cfg file with the new "cerdType" parameter if it is not present (if "cerdType" parameter is present, it does not add or change it):

```
# Type of (Universal) Combined Exciter Receiver
# Data Acquisition Board (CERD)
# type = integer
# Usage
# X Y Y
# \ \ \----- (U) CERD Slot #1 (VME slot 13 , right next to
# power supply)
# \ \----- (U) CERD Slot #2 (VME slot 10)
# \----- MKE or YMS Receiver mapping
# Where X =
#   0: MKE receiver mapping scheme
#   1: YMS receiver mapping scheme
#
# Where Y =
#
#   0 : No(U) CERD Installed
#   1 : Narrowband CERD
#   2 : Universal CERD (UCERD)
# Legal Combinations
#
#   0yy: MKE default receiver mapping
#   1yy: YMS default receiver mapping
#   x01: Narrowband CERD on Slot #1
#   x02: Universal CERD on Slot #1
#   x21: Narrowband CERD on Slot #1 ;Universal CERD on Slot #2
#   x22: Universal CERD in both Slot #1 and Slot #2
cerdType = "001"
```

3. The "configup" script updates the GradientConfig.cfg file with the new Full Power SGD parameters if not present (if the following parameters are present, it does not add or change them):

```
# Added by shivar@wge for MRIge49945
Sdlim = "0.60"
Lcoil = "1.40"
Rac = "152"
Pmgs = "3000"
mGRAMpower = "9000"
```

REVISION HISTORY

[illegible]